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A scoping literature review of prompt engineering for bridging students AI literacy in higher education

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A B S T R A C T

Generative Artificial Intelligence (GenAI) has been widely used in education, yet many students still lack the skills to fully leverage its potential. A systematic search was conducted across three academic databases from 2020 to 2025. This scoping review explores the evolving landscape of GenAI applications and introduces prompt engineering as a critical competency students need to develop to bridge their AI literacy and enhance their technical skills and critical thinking in higher education. A search across Scopus, PubMed, and ERIC yielded 76 studies that were systematically reviewed to identify relevant research. The PRISMA-ScR guidelines were used to determine which studies met the eligibility requirements. The main findings of this scoping review were to map the definitions and applications of prompt engineering in higher education, the prompt strategies to bridge students' AI literacy, the implementation methods, the types of student tasks, the research findings, and the research gaps. Thematic analysis yielded six key themes: prompt engineering, AI literacy, guided learning design, performance and learning outcomes, student behavior, and GenAI usage risks, all of which are critical areas to consider in optimizing GenAI usage. This review identifies critical gaps in current understanding, particularly regarding prompt engineering to maximize GenAI capabilities across diverse student conditions and curriculum areas and to bridge AI literacy. The findings provide a preliminary contribution to lecturers, GenAI tool developers, researchers, and policymakers by offering direction for curriculum integration, tool design, and future empirical research on prompt engineering to strengthen student AI literacy in higher education.

1. Introduction

The integration of Generative Artificial Intelligence (GenAI) technology in higher education has changed how students learn, including in writing and problem-solving activities (Su et al., 2023; Urban et al., 2024). This integration is the latest development in the broader vision of applying artificial intelligence in education (AIED). Hwang et al. (2020) define AIED as the use of AI technology in educational environments to facilitate teaching, learning, or decision-making, with the main goal of providing personalized learning guidance for each learner. Hwang et al. (2020) emphasize that the interdisciplinary nature of AIED presents unique challenges, as without understanding the role and workings of AI technology, researchers and practitioners risk failing to implement AIED applications effectively. These interdisciplinary challenges are becoming increasingly relevant as students now interact directly with AI systems through prompts, requiring specific skills to fully utilize the potential of GenAI. However, in practice, many students still lack the skills to fully utilize the potential of GenAI (Kim et al., 2025). UNESCO (2023) findings state that most higher education institutions do not yet have a formal curriculum framework for AI literacy, so students tend to use GenAI self-taught without an in-depth understanding. Students often perform unskilled prompting when interacting with GenAI, resulting in inaccurate outputs (Knoth et al., 2024). To fully utilize the potential of GenAI, a careful strategy is needed to formulate and optimize prompt inputs (L. Wang, Chen, Deng, et al., 2024). Input prompts are instructions given to GenAI to perform tasks according to the user's wishes. There is a term that describes the strategy of formulating input prompts for GenAI called prompt engineering (Zaghir et al., 2024). Prompt engineering in the context of education has gained recognition from the international academic community. Allison et al. (2025), in a multidisciplinary editorial involving 14 experts in educational computing, explicitly identified prompt engineering as one of the recommended

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research focuses, namely designing tasks and strategies through the effective use of GenAI to promote students' computational thinking and problem-solving skills. This recognition confirms that prompt engineering goes beyond mere technical skills and has been positioned as an educational computing practice that needs to receive systematic research attention. Prompt engineering is a skill that students must pay attention to because, through understanding the principles of prompt engineering, students can optimize their interaction with GenAI while developing critical thinking and AI literacy skills (Knoth et al., 2024; Lo, 2023). AI literacy is a competency that enables users to communicate and collaborate effectively with GenAI and critically evaluate GenAI outputs (Long & Magerko, 2020). Therefore, the use of prompt engineering helps students understand and apply AI literacy to be critical of the use of GenAI responses (Walter, 2024). The relevance of this study is further strengthened by the transformation of the role of students due to the presence of GenAI in various educational purposes, which also raises important issues regarding the right of students to learn GenAI (Lai & Tu, 2024).

The role of prompt engineering in bridging student AI literacy has attracted considerable research interest. Researchers examined how students' prompt engineering skills developed dynamically alongside their GenAI literacy in higher education, showing that GenAI-related knowledge, skills, and attitudes developed mutually through continuous practical engagement (Yan et al., 2025). In line with this, the specific use of prompt engineering in academic writing can influence students' AI literacy. Students with high AI literacy tend to be able to collaborate with GenAI and obtain high scores in terms of content, structure, and writing, compared to students with low AI literacy, who only use it for ideation (Kim, Yu, et al., 2025). The use of prompt engineering in GenAI can also improve students' learning outcomes and critical thinking skills in response to GenAI (Gong et al., 2025). The critical thinking skills developed by students include an understanding of the evaluation dimension of AI literacy (Wang et al., 2023). The benefits of using prompt engineering by students can also reduce their anxiety about GenAI technology, making students more confident in integrating GenAI into their learning tasks (Davila-Moran et al., 2025). Furthermore, studies show that the use of GenAI without the support of prompt engineering has the potential to produce inaccurate GenAI responses and cause negative impacts, especially for students who do not yet have adequate AI literacy, because prompts that are still general in nature can trigger inappropriate outputs or even contain fictitious information that misleads students' understanding (Güner & Er, 2025). This risk is in line with the concerns identified by experts in educational computing. Hwang warns that students may become too dependent on GenAI assistance to the point of ceasing to attempt to complete tasks independently, a condition referred to as metacognitive laziness (Allison et al., 2025). This dependence reinforces the urgency of developing prompt engineering as a skill that ensures students remain cognitively engaged when interacting with GenAI. This strengthens the argument that prompt engineering needs to be formally recognized as an essential 21st-century skill within the framework of higher education competencies (Federiakin et al., 2024).

Several empirical studies exploring the relationship between prompt engineering and student AI literacy in higher education are still limited, and the findings are scattered. As a result, a comprehensive synthesis of these studies is not yet available. To address this gap, a scoping review is needed. Previous reviews have discussed the mapping of *prompt engineering* research for medical applications (Zaghir et al., 2024). This is very different from the context of this review, which raises the mapping of research on *prompt engineering* to bridge student AI literacy in higher education. This *scoping review* selected three databases to provide an in-depth examination of the results of studies focusing on *prompt engineering* to bridge student AI literacy. This scoping review not only identifies existing research gaps but also provides valuable preliminary insights for lecturers as learning facilitators in encouraging the use of prompt engineering when interacting with GenAI in higher education. The use of prompt engineering is intended to facilitate students' AI literacy in using GenAI while being aware of the risks involved, skilled in producing targeted GenAI output, and using GenAI output responsibly. This review also aims to map the relationship between the prompt engineering used by students and the dimensions of AI literacy that emerge in the classroom, so that lecturers have practical references in designing learning activities, assessment instruments, and feedback schemes.

This scoping review raises the following research questions.

1) How does the literature define and apply prompt engineering in higher education?
2) What prompt strategies have been reported to bridge student AI literacy in higher education learning, and what types of GenAI are used?
3) What research methods are used to apply prompt engineering in student learning?
4) In what types of tasks is prompt engineering used in higher education?
5) What research gaps still exist in prompt engineering research?
6) What are the results of research interventions involving prompt engineering to bridge the AI literacy gap among higher education students?

2. Literature review

2.1. Prompt engineering

GenAI can generate or modify its text output based on the input text (White et al., 2023). A high-quality prompt essentially has a structure that directs GenAI on which information is important so that it produces the output desired by the user (White et al., 2023). There are various prompting techniques, from basic to complex approaches, to improve GenAI capabilities (Sahoo et al., 2024). Zero-shot learning involves providing prompts to GenAI without any examples, but it can be improved through refining instructions and reinforcement learning from human feedback. (Dang et al., 2022). Few-shot learning involves providing prompts that instruct GenAI to learn new tasks with few examples so that it can submit tasks directly and improve GenAI performance (Ma et al., 2023). For users who are not experts in interacting with GenAI, the use of prompt engineering can help direct GenAI responses towards the results desired by users (Yang et al., 2025). More relevant outputs occur when combining the few-shot learning technique with chain-of-thought, which involves creating intermediate natural language reasoning steps to guide GenAI in learning challenging tasks (Wei et al., 2022).

There are developments in elements to facilitate the prompt engineering process to direct highly relevant outputs from GenAI, which recommends six elements in prompt composition, namely role setting, task, context, format, example, and tone (Murray, 2025). These six elements can be simplified into four main components, namely role assignment, context provision, boundary determination, and result verification. This simplification is done because the elements of task, format, example, and tone basically function as a form of contextualization and limitation that directs GenAI output (Federiakin et al., 2024). Verification of results is added as an explicit element given GenAI's tendency to produce information that appears convincing but is inaccurate, known as hallucination (Ji et al., 2023). Result verification is a critical component because, without a checking mechanism, users risk accepting false outputs as facts, especially in academic contexts that demand high accuracy (Alkaissi & McFarlane, 2023).

Various techniques and elements of prompts also influence how prompt engineering is defined in the literature. This is because prompt engineering is still experiencing terminological conflicts and fragmented ontological understanding due to its relatively recent emergence (Schulhoff et al., 2024). Some studies explicitly define prompt engineering as a standalone technical procedure, complete with prompt characteristics, assessment rubrics, and interaction log monitoring, while others define it implicitly, merging the skill of composing prompts into the broader competency of GenAI use without providing a separate operational definition. This difference in definitional approaches is not a weakness of the literature, but rather reflects the nature of a field that is still evolving. This definitional variation can be understood through the perspective of the philosophy of science. Giovannini and Schiemer (2021) explain that explicit definitions logically equate new concepts with formulas in the base language so that the concepts can be eliminated without changing existing theorems, while implicit definitions interpret a term through unique conditions within a theoretical model defined by axioms, with meanings determined contextually in relation to other basic terms. In first-order logic, Beth's Theorem states that every implicit definition is ultimately equivalent to an explicit definition (Giovannini & Schiemer, 2021).

These definitional differences have different pedagogical implications. Explicit definitions lead to specially structured training designs, such as *prompt* writing workshops with separate modules and trained instructors, while implicit definitions allow for the integration of *prompt* skills into existing courses without the need for separate time allocations (Ellis, 2009; Rebuschat & Williams, 2012). These two definitional approaches complement each other: the explicit approach provides operational clarity for teaching, training, and assessment practices, while the implicit approach ensures cross-course connectivity and strengthens ethics-based learning governance. Within the framework of this review, this categorization will be used as an analytical lens in the Results section to map how the reviewed studies position *prompt engineering* in the context of higher education.

2.2. Student AI literacy

The rapid development of AI technology requires humans to have the ability to interact with this technology independently and skillfully (Carolus et al., 2023). This ability can be developed through AI literacy, which enables users to interact with AI technology safely and effectively. Wang et al. (2023) define AI literacy as the ability to recognize and understand AI technology in practical applications; the ability to apply and utilize AI technology to complete tasks efficiently; the ability to critically analyze, select, and evaluate data and information provided by AI; and the development of awareness of personal responsibility and respect for mutual rights and obligations. This definition shows that AI literacy covers a broad dimension, requiring a conceptual framework that can translate it into more structured competencies that can be operationalized in educational practice.

Various efforts have been made to conceptualize AI literacy. Long and Magerko (2020) pioneered an AI literacy framework through an interdisciplinary review that resulted in 17 competencies in five themes, namely: *what is AI?, what can AI do?, how does AI work?, how should AI be used?*, and *how do people perceive AI?* This framework provides a comprehensive conceptual foundation, but it is more oriented towards a technical understanding of the internal mechanisms of AI, such as knowledge representation and machine learning steps, and therefore does not operationalize user competencies in interacting with prompt-based GenAI. Furthermore, Ng et al. (2021) proposed four aspects of AI literacy based on an adaptation of Bloom's Taxonomy, namely: *know and understand, use and apply, evaluate and create*, and *ethical issues*. This framework makes an important pedagogical contribution, but the *evaluate and create* dimension covers the ability to build and create AI applications that are more technical in nature than user literacy. Furthermore, the framework was developed before the GenAI era and is primarily intended for K-12 contexts, so it does not yet accommodate interactions with prompt-based GenAI, which is a key characteristic of GenAI use by students in higher education.

Unlike the two frameworks above, Wang et al. (2023) developed an AI literacy framework that specifically focuses on user competence in using AI technology. This framework is based on an analogy with digital literacy, specifically the *technological-cognitive-ethical* model (Calvani et al., 2008) and the KSAVE (*Knowledge, Skills, Attitudes, Values and Ethics*) framework (Wilson et al., 2015), which was then derived into four psychometrically validated constructs through a 12-item scale on two independent data samples, namely: awareness, usage, evaluation, and ethics. The results of confirmatory factor analysis show that this four-construct model is the most adequate representation of AI literacy.

The AI literacy framework from Wang et al. (2023) was selected in this review for three main reasons. First, this framework focuses on user competencies in identifying, using, evaluating, and considering ethics when interacting with AI technology rather than on understanding the internal mechanisms of AI, making it most suitable for analyzing how students interact with GenAI through prompt engineering. Second, this framework has empirically confirmed validity and reliability, providing a reliable conceptual basis for mapping the findings of this scoping review. Third, these four parsimonious constructs can be directly mapped onto student activities when performing prompt engineering: awareness is evident when students understand the capabilities and limitations of GenAI; use is evident when students skillfully compose prompts; evaluation is evident when students verify GenAI outputs; and ethics are evident when students consider academic integrity and responsibility of use.

AI literacy is a crucial factor in education, especially in higher education (Crompton & Burke, 2023). With the expected significant transformation in higher education due to the ease of access to GenAI, it is important for students to understand, use, and evaluate GenAI outputs to ensure safe interaction (Bozkurt, 2023; Kasneci et al., 2023). The urgency of AI literacy aligns with the warning from Hwang et al. (2020), who emphasized that without understanding the role and workings of AI technology, researchers and practitioners will find it difficult to implement AIED applications effectively. This warning has become even more concrete in the era of GenAI, where students are no longer merely passive users of AI systems, but interact directly through prompts that determine the quality of the output. Thus, AI literacy is not just an additional competency, but a prerequisite for students to be able to utilize the role of GenAI as an intelligent learning tool productively and responsibly.

2.3. Prompt engineering to bridge student AI literacy

Prompt engineering has elements to direct relevant outputs from GenAI, consisting of role assignment, context, limitations, and verification (Murray, 2025). These components can indirectly help students understand the four dimensions of AI literacy, namely awareness, use, evaluation, and ethics. (Wang et al., 2023). Although the two are interrelated, it is important to emphasize that prompt engineering and AI literacy are two conceptually different constructs. As defined by Wang et al. (2023), AI literacy is a competency framework that identifies what users of AI technology must master. This framework is rooted in the tradition of digital literacy, particularly the *technological-cognitive-ethical* model (Calvani et al., 2008), which is declarative in describing the competencies that need to be possessed. Rather than being a mere subset of AI literacy, prompt engineering is a structured practice that has its own procedural components of role assignment, context, constraints, and verification and has been positioned by recent literature as a standalone 21st-century skill (Federiakin et al., 2024), Therefore, prompt engineering in this review is not positioned as a subset of AI literacy, but rather as an operational mechanism that shows how AI literacy competencies are activated in the real-world practice of student interaction with GenAI.

The functional relationship between the components of prompt engineering and the dimensions of AI literacy can be mapped specifically to student practice in the classroom. The awareness dimension is evident when students are aware of the capabilities, risks, and appropriate timing of using GenAI in higher education (Knoth et al., 2024); in prompt engineering practice, this dimension is activated when students define roles and formulate contexts in prompts because the process requires an understanding of what GenAI can and cannot do. The use dimension is evident when students operate GenAI skillfully (Caner-Yıldırım, 2025; Hwang et al., 2025; Knoth et al., 2024). This dimension is activated when students compose prompts in a structured manner with clear boundaries and manage multi-turn conversations iteratively. The evaluation dimension is evident when students enter prompts to verify the output (Aruleba et al., 2025; Huang et al., 2024; Knoth et al., 2024; Picardal, 2025); This dimension is activated when students request step-by-step explanations and verify outputs against reliable sources. The ethics dimension encompasses students' attention to academic integrity or awareness of bias in GenAI outputs (Aruleba et al., 2025; Caner-Yıldırım, 2025); this dimension is activated when students consider academic integrity, proper attribution, and potential bias when composing and evaluating prompts. This correspondence pattern is supported by the findings of Knoth et al. (2024), which show that aspects of AI literacy correlate with the quality of students' *prompt engineering* practices, as well as by Yan et al. (2025), who document the dynamic relationship between the two constructs. Thus, *prompt engineering* serves as an operational vehicle that transforms AI literacy competencies from the conceptual level into observable practices in the classroom.

Although the mapping above shows that the right prompt engineering can produce high-quality GenAI output and strengthen students' AI literacy. (Knoth et al., 2024 High-quality output alone is not enough to measure the success of GenAI integration in learning. Huang (in Allison et al., 2025) warns that research on educational computing tends to focus too much on easily measurable outcome metrics, such as test scores and completion rates, while neglecting students' thinking processes. This warning becomes even more relevant when GenAI can produce sophisticated outputs with minimal cognitive investment. Therefore, mapping prompt engineering to AI literacy must include not only output results, but also students' cognitive processes in designing, evaluating, and verifying prompts, which is the core of AI literacy itself.

3. Method

This scoping review was conducted and reported in accordance with the PRISMA guidelines for scoping reviews (PRISMA-ScR) (Tricco et al., 2018). The PRISMA-ScR guidelines provide a standard methodology for conducting scoping reviews, which are useful for mapping the literature in emerging fields with developing evidence.

3.1. Literature search and selection criteria

Inclusion and exclusion criteria were established prior to the screening process to ensure that the selected articles were suitable for effectively answering the research questions. The inclusion criteria required that articles be empirical studies published in English that focused on discussing *prompt engineering* strategies to bridge the AI literacy gap among college students and had undergone a rigorous peer review process. Conversely, exclusion criteria apply to non-empirical articles, such as conceptual articles, theoretical articles, review articles, and editorials, articles not published in English, articles that do not focus on discussing *prompt engineering* strategies to bridge AI literacy among university students, and articles that have not undergone a rigorous peer review process. In this scoping review, information sources were selected from three databases, namely Scopus, ERIC, and PubMed. The selection of articles is limited to those published between 2020 and October 9, 2025. The selection of articles published since 2020 reflects the beginning of extensive research on GenAI. This is related to the development of *prompt engineering* publications as a tool for instructing the use of GenAI (Liu et al., 2023).

3.2. Description of the search process

The search string was developed with the help of ChatGPT to generate a list of keywords relevant to the topic. The search was conducted to identify relevant studies by logically combining various keyword variations, namely “generative AI” and “prompt” as well as “AI literacy” and “higher education” using the Boolean operators ‘AND’ and “OR.” An initial search was conducted in each database to assess the relevance of the initial search results. After reviewing the search results in each database, the search string was refined. Three databases used the following search string: (“generative AI” OR “large language models”) AND (“prompt engineering” OR ‘prompt’) AND (“AI literacy” OR “artificial intelligence literacy”) AND (“higher education” OR “university”). The search results using this string yielded a total of 76 articles from three databases, indicating that research in this field is growing. After removing one duplicate, 75 articles remained for further screening.

3.3. Literature screening process

To ensure the suitability of the selected articles and effectively answer the research questions, a set of inclusion and exclusion criteria was established prior to the screening process. Based on these criteria, the screening process was conducted in two stages. In the first screening stage, the titles and abstracts of all articles were reviewed, resulting in 58 articles being excluded and 17 articles being saved for review of the full text of all remaining articles. The remaining 17 articles were accessible in full text and could proceed to the second screening stage. In the second screening stage, the entire text structure of each remaining article was reviewed, resulting in the exclusion of 2 non-empirical articles, leaving 15 articles that met the criteria. Finally, 15 articles were included in the final review. The number of 15 studies that met the inclusion criteria reflects the nature of a developing field, given that empirical studies on *prompt engineering* for AI literacy in higher education only began to emerge significantly after the launch of ChatGPT in November 2022 (Tu & Lu, 2025).

This number is in line with common findings in scoping reviews in emerging fields, where the number of available studies tends to be small, as emphasized by Arksey and O'Malley (2005) that scoping reviews are designed to map the breadth of available evidence regardless of the number of studies found. The PRISMA-ScR flow diagram illustrating the literature search and screening process is presented in Fig. 1.

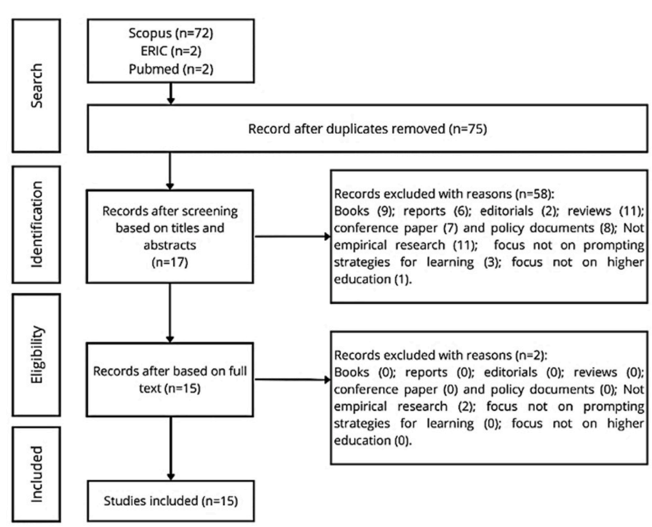

Fig. 1. PRISMA-ScR flow diagram.

3.4. Data coding and analysis

The coding and analysis process was carried out in accordance with the research questions. Research question 1 (RQ1) discusses the definition and application of prompt engineering in higher education. Research question 1 relates to the data extraction results, including authors, years, definitions, and the application of prompts in higher education. Research question 2 (RQ2) discusses the prompt strategies reported to bridge student AI literacy. Research question 2 relates to the coding categories of the extracted data results, including the prompt strategies used in learning, the AI literacy introduced to students, and the types of GenAI used. Research question 3 (RQ3) discusses the methods used in the study. Research question 3 relates to the coding categories of the extracted data results, including the research methods used. Research question 4 (RQ4) discusses the context of task types for the use of prompt engineering in higher education. Research question 4 relates to the coding category of extracted data results, including the context of task types. Research question 5 (RQ5) discusses the existing research gaps in the application of prompt engineering in higher education. Research question 5 relates to the coding category of extracted data results, including research gaps.

The sixth research question (RQ6) used thematic analysis. This process was carried out through the following steps. First, the research intervention results from each article were collected. Second, for each intervention result or main finding, keywords were identified, such as: "student prompting strategies," "dimensions of student AI literacy," and "student self-efficacy." Third, the results from the identification of keywords and findings were grouped into main themes and sub-themes. Fourth, for each identified noun phrase, coding was performed by retaining the original noun phrase or replacing it with a closely related term. If a newly identified noun phrase was similar to a previously coded phrase, it was given the same code; if there was no code closely related to the one previously assigned, a new code was introduced.

An illustration of the thematic analysis performed is the research results from Andewi et al. (2025), which state that the group that used ChatGPT with a gradual prompting strategy showed a higher increase in writing skills, thereby increasing self-efficacy compared to the group that did not use GenAI. Two keywords were identified, namely "gradual prompting strategy" and "student self-efficacy." The noun phrase "student self-efficacy" is closely related to "student self-confidence," which has been used in previously coded studies, and many of the studies reviewed refer to "self-confidence." Therefore, the phrase "student self-efficacy" was coded as "student self-confidence".

During the coding and analysis process, the first author and second supervisor independently coded the first seven articles as a calibration phase. The two researchers then compared their coding results. Next, the two researchers independently continued coding the remaining eight articles. Inter-coder reliability was measured using Cohen's Kappa (Cohen, 1960) on all 15 articles. The results showed a κ coefficient of 0.775 ($p < 0.001$), indicating a substantial level of agreement based on the interpretation guidelines of Landis and Koch (1977). After all themes were established, the second supervisor verified the overall coding results to ensure consistency and accuracy of the analysis. Any disagreements that arose during the verification process were resolved through additional consensus discussions.

In accordance with the PRISMA-ScR guidelines (Tricco et al., 2018), a formal critical appraisal was not conducted in this scoping review because its main objective was to map the breadth of evidence available in the field of prompt engineering for AI literacy in higher education, rather than to evaluate the quality of individual pieces of evidence. Nevertheless, to enhance methodological transparency, Table 3 presents a descriptive summary of the methodological characteristics of each included study, including the type of evidence, data collection instruments, and study duration, in addition to information on prompt strategies and AI literacy.

4. Result

4.1. Definition and application of prompt engineering in higher education

Table 1 maps prompt engineering into two complementary conceptual approaches. Prompt engineering is defined explicitly and implicitly in the literature. Prompt engineering is explicitly defined as the process of designing, formulating, and optimizing prompts to be entered into GenAI with the aim of producing desired, accurate, and contextually appropriate outputs. Meanwhile, prompt engineering is implicitly defined as part of GenAI usage competencies within the AI literacy framework integrated into the education curriculum. Based on Table 1, there are 10 studies that explicitly define prompt engineering, conducted by researchers from countries including Indonesia, the United Kingdom, Chile, France, China, South Korea, Germany, the Philippines, the United Arab Emirates, and Australia. Meanwhile, there are 5 studies that implicitly define prompt engineering, from researchers from Turkey, Australia, China, the United Kingdom, and Spain. The categorization of the definitions of *prompt engineering* into explicit and implicit emerged inductively during the data analysis process. This inductive approach is consistent with the purpose of a *scoping review*, which is designed to identify patterns from the available evidence in a developing field (Tricco et al., 2018). This categorization is grounded theoretically in the philosophy of definition of Giovannini and Schiemer (2021) explaining that explicit definitions equate new concepts directly with formulas in basic language, while implicit definitions interpret terms through conditions of unique interpretation in theoretical models; Beth's theorem establishes that both are deductively equivalent. This definitional difference is not merely terminological, but has direct implications for instructional design or how to teach *prompt engineering*, as summarized in Table 2.

Table 1 shows the number of applications of prompt engineering in learning, starting from programming taught to students through step-by-step prompt sheets to build text-based games using Python (Gong et al., 2025). Meanwhile, in design studios, prompt engineering is applied as a core competency that guides the exploration of ideas, the evaluation of GenAI outputs, and iterative refinement to find the right GenAI outputs (Fleischmann, 2024). The application of prompt engineering in English language learning is carried out with students to practice composing effective prompts while evaluating GenAI outputs (Partridge et al., 2024). Meanwhile, the application of prompt engineering in research methodology is systematically documented through a reflection journal containing hundreds of prompts (Hwang et al., 2025). In the application of obstetrics and gynecology learning, student interactions are mapped into types of questions to formulate recommendations for strengthening AI literacy (Desseauve et al., 2024).

These two definitional approaches are complementary and each contributes differently to the practice of prompt engineering in higher education. The explicit approach provides operational clarity because prompt engineering is defined as a distinct skill that can be trained through specialized modules, assessed using standardized rubrics, and monitored through interaction logs (Carrasco-Sáez et al., 2025; Knoth et al., 2024). This approach aligns with the principle of explicit learning, which is the process of acquiring knowledge that is conscious and can be explained verbally through formal instruction (Kim & Godfroid, 2023). In contrast, the implicit approach ensures cross-course connectivity by integrating *prompt* writing skills into existing learning activities without requiring separate time allocation (Caner-Yıldırım, 2025; Fleischmann, 2024), while also reinforcing the ethical governance of GenAI use. This approach is consistent with implicit learning, where knowledge of patterns or rules is acquired unconsciously but reflected in automatic performance (Weinberger & Green, 2022). Practically, the complementarity of these two approaches provides flexibility for instructors to choose or combine them according to the course context, student characteristics, and learning objectives to be achieved.

Prompt engineering has been widely applied in various higher

Table 1
Definition and application of *prompt engineering* in higher education.

Author (Year)	Country	Definition of Prompt Engineering	Application of Prompt Engineering
Andewi et al. (2025)	Indonesia	Prompt engineering is positioned as a strategy for developing increasingly targeted prompts through gradual conversation to improve output quality.	This study applies prompt engineering in the context of English language learning among students to compare the use of GenAI with faculty guidance while practicing prompt composition.
Aruleba et al. (2025)	United Kingdom	Prompt engineering is operationalized as the development and iteration of prompt composition strategies to make outputs more accurate and contextually appropriate.	This study applies prompt engineering in the context of computer science classes for students through the completion of five programming assignments accompanied by reflective journals and interviews, as well as training in prompt composition and critical assessment of output.
Caner-Yıldırım (2025)	Turkey	Prompt engineering is implicitly defined as the skill of composing prompts, which is part of GenAI competency.	This study applies prompt engineering in the context of curriculum development in higher education by guiding students to master instruction writing skills as part of GenAI competencies.
Carrasco-Sáez et al. (2025)	Chile	Prompt engineering is defined as the process of compiling clear, effective, relevant, and contextual instructions to produce the desired output.	This study applies prompt engineering to academic tasks by placing lecture materials and sample assignments as context for students to design prompts based on specific objectives to ensure alignment with learning outcomes, assessment, and assignment constraints.
Desseauve et al. (2024)	Swiss and France	Prompt engineering is defined as a new skill for utilizing GenAI.	This study applies prompt engineering in the context of medical education through observation of interactions with GenAI by obstetrics and gynecology students, resulting in the grouping of question types and recommendations for AI literacy programs.
Fleischmann (2024)	Australia	Prompt engineering is implicitly defined as an element of GenAI curriculum design.	This study applies prompt engineering in the context of curriculum development, which places prompt writing as a core competency in design studios so that students learn to write contextual prompts to explore ideas, evaluate outputs, and iteratively refine them in search of the right Gen-AI outputs.
Gong et al. (2025)	China	Prompt engineering is defined as the activity of designing, composing, and optimizing inputs so that GenAI produces specific responses with the aim of improving the quality of human-computer interaction through carefully composed prompts.	This study applies prompt engineering in the context of programming education in higher education through progressive prompt sheets that combine general prompts and advanced prompts to guide students in building text-based adventure games in Python for 60 min with the support of artificial intelligence assistants.
Huang et al. (2024)	China	Prompt engineering is implicitly defined as the skill of composing and inputting prompts into generative artificial intelligence systems to respond appropriately to context.	This study applies prompt engineering in the context of formative assessment in an eight-week undergraduate research methods course by examining the patterns of student prompt composition and how they evaluate GenAI outputs.
Hwang et al. (2025)	South Korea	optimizing prompts so that GenAI produces the desired responses, which includes response control, language selection, contextualization, and format composition in multi-turn interactions.	This study applied prompt engineering in the context of a research methodology course at a university with 10 EFL graduate students who used GenAI for reaction paper assignments and recorded 238 prompts in their reflection logs.
Knoth et al. (2024)	Germany	Prompt engineering is defined as the skill of composing precise and structured prompts to trigger the desired output.	This study applies prompt engineering in the context of evaluating students' skills in composing prompts for travel planning assignments and research projects that assess the quality of prompts and outputs through integrative complexity scores.
Partridge et al. (2024)	United Kingdom	Prompt engineering is implicitly positioned as the skill of creating effective prompts within the framework of AI literacy.	This study applies prompt engineering in the context of English preparation programs for academic purposes by training students to write effective prompts and evaluate GenAI output in class.
Picardal (2025)	Philippines	Prompt engineering is defined as the use of academically designed smart prompts to encourage original ideas and critical thinking.	This study applies prompt engineering in the context of academic writing courses for prospective science teachers by using

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Table 1 (*continued*)

Author (Year)	Country	Definition of Prompt Engineering	Application of Prompt Engineering
			smart prompts as step-by-step guides and feedback tools.
Rovira-Esteva (2025)	Spain	Prompt engineering is implicitly defined as part of AI literacy in order to have ethics in using GenAI.	This study applies prompt engineering in the context of first-year learning in higher education by guiding students in the use of GenAI with regard to ethics and pedagogy.
Sawalha et al. (2024)	United Arab Emirates	Prompt engineering is operationally defined through mapping the characteristics and qualities of prompts that influence GenAI output.	This study applies prompt engineering in the context of embedded systems courses in higher education by analyzing student prompt logs and GenAI model outputs to show that clearer and more contextual prompts can be associated with more accurate task completion.
Quince and Nikolic (2025)	Australia	Prompt engineering is operationally defined as the introduction of prompt composition techniques to generate output and evaluate quality, bias and ethical considerations.	This study applied prompt engineering through structured training to engineering students, and the training group showed higher learning outcomes than the untrained group and the control group, thereby strengthening the students AI literacy.

Table 2
Implications of teaching *prompt engineering* from explicit and implicit categorization.

Aspect	Explicit	Implicit
Definition	Taught as a separate skill with clear technical procedures	Embedded in other tasks/ activities without being defined separately.
Example	Workshop on prompt development with modules, rubrics, and interaction logs (Carrasco-Sáez et al., 2025; Knoth et al., 2024).	Research tasks using GenAI with specific guidance as part of broader competencies (Caner-Yıldırım, 2025; Fleischmann, 2024).
Resources	Requires dedicated time, trained instructors, and specific modules	Can be directly integrated into existing courses
Theoretical basis	Explicit learning: conscious knowledge, acquired through formal instruction (Kim & Godfroid, 2023).	Implicit learning: unconscious knowledge, reflected in automatic performance (Weinberger & Green, 2022).

education learning contexts and strengthens the learning process in various disciplines. Prompt engineering is often used in English language learning, appearing in four studies, and computer science, appearing in four studies. This is followed by its application in engineering learning in two studies. The application of prompt engineering in other learning areas includes design, economics, obstetrics and gynecology, psychology, and translation. Prompt engineering is widely applied in English language learning, particularly in writing activities, as it is highly relevant to prompt strategies that are repetitive and easy to teach, document, and assess (Wang & Dang, 2024). Prompt engineering is also widely applied in computer science learning in higher education because GenAI improves the efficiency of programming work, including code creation and explanation, debugging, and providing quick feedback. However, its performance is highly dependent on the quality of the prompt provided, making prompt engineering an important prerequisite for successful computational learning (Agbo et al., 2025).

The annual publication trend of empirical studies on prompt engineering over the past few years shows a clear upward trend in 2024-2025. Based on the articles included in this review, there were 6 studies published in 2024, which then increased to 9 studies in 2025. This phenomenon correlates temporally with the massive adoption of ChatGPT following its launch in November 2022, which created a pedagogical urgency to understand how students can effectively utilize GenAI through prompt engineering skills. This surge in publications signals an academic response to the urgent need to bridge student AI literacy amid rapid technological transformation. This growth trend is expected to continue through 2026. This projection is supported by the fact that the number of relevant publications available in the database shows a significant increase towards the end of 2025. However, given that the number of studies included in this review is still limited, the reported distribution of *prompt engineering* applications across disciplines should be understood as preliminary indications and cannot be generalized without broader evidence.

4.2. Prompt strategies to bridge student AI literacy

Table 3 shows the *prompt* strategies used to bridge student AI literacy in learning, starting from role assignment, formulating the desired context and tasks, setting boundaries, and requesting explanations or verification. Researchers in every study often pay attention to the *prompt* strategy of requesting explanations or verification. This is because GenAI systems can sometimes be less transparent and can produce responses that are not as desired, influenced by the level of hallucination (Andewi et al., 2025). Based on these uncertain responses, students are encouraged to verify and assess the reliability of the outputs generated by GenAI (Rapp et al., 2025).

The use of prompt strategies in learning can provide an understanding of AI literacy. AI literacy has four dimensions, namely the awareness dimension, the usage dimension, the evaluation dimension, and the ethics dimension (Wang et al., 2023). Based on the summary in Table 3, 11 of the 15 studies included introduced students to the dimension of awareness through prompt engineering (73%).

Next, 14 of the 15 studies included provided students with an introduction to the dimensions of use and evaluation through prompt engineering (93%), and only 8 of the 15 studies included provided students with an introduction to the ethical dimension through prompt engineering (53%). The finding of a low ethical dimension compared to other AI literacy dimensions suggests that the ethical dimension still requires more systematic integration through learning design and policies in higher education (Dabis & Csáki, 2024).

Prompt engineering aims to direct GenAI to produce outputs that are appropriate to the context, but one dimension of AI literacy, namely the awareness dimension, emphasizes the importance of choosing the type of GenAI according to the needs (Aruleba et al., 2025; Knoth et al., 2024). Based on the recap in Table 3, the type of GenAI that is often used is ChatGPT, which appears in 14 studies. Other types of GenAI are also mentioned, including Midjourney, Adobe Firefly, DALL-E, Bing Image Creator, Gemini, and Perplexity AI. The dominance of ChatGPT in the reviewed corpus may reflect the availability and popularity of this tool at the time of the research, compared to other GenAI.

4.3. Methods for applying prompt engineering in student assignments

Table 3 shows that the majority of included studies used qualitative or mixed designs with relatively small sample sizes. Of the 15 studies, 7

Table 3
Summary of included studies: prompt strategies, AI literacy, and methodological characteristics.

Author (Year)	Population	Types of GenAI	Prompt Strategy	AI literacy introduced to students	Method	Type of Evidence	Instrument	Duration	Task Type	Research Gap
Andewi et al. (2025)	65 English language students	ChatGPT	The prompt strategy used includes multi-turns with role and context setting, requests for clarification and verification.	Dimensions introduced to students indirectly include the dimensions of awareness, usage, evaluation and academic ethics.	Mixed method	Mixed	Test and interview	8 weeks	Academic writing in English	There is insufficient evidence regarding the effectiveness of prompt design compared to lecturer interaction, and there is a lack of understanding of the development of student prompt design strategies in English as a Foreign Language learning.
Aruleba et al. (2025)	21 computer science students	ChatGPT	The prompt strategy used emphasizes iterative formulation with positive and negative examples and gradual improvement accompanied by verification.	Dimensions introduced to students indirectly include the dimension of awareness, usage and evaluation.	Qualitative method	Self-reported	Reflective journal and interviews	4 weeks	Computer programming assignments	There is no long-term study that tracks the development of prompt design strategies, critical evaluation and ethical reflection among computer science students in authentic programming tasks.
Caner-Yıldırım (2025)	404 Turkish students	ChatGPT	The prompt strategy used involves managing multi-turn dialogues with GenAI and critically evaluating AI output.	Dimensions introduced to students indirectly include the dimensions of usage and evaluation.	Quantitative method	Self-reported	questionnaire	Cross-sectional	General course activities	Limited understanding of GenAI adoption factors in higher education and the need to update the digital competency framework to include prompt engineering and AI literacy for GenAI.
Carrasco-Sáez et al. (2025)	102 Chilean students	ChatGPT	The prompt strategy used focuses on contextualizing the prompt, formulating specific objectives, setting domain boundaries and verifying the output of GenAI.	Dimensions introduced to students indirectly include the dimensions of awareness, usage and evaluation.	Structured exploratory study	Mixed	AI interaction log	1 session	Spanish language and text-based reasoning assignments	There is little empirical evidence on prompt design strategies other than English, and it is unclear which techniques most improve the quality of GenAI responses.
Desseauve et al. (2024)	Swiss medical student	ChatGPT	The prompt strategy used guides the formulation of clinical prompts, verification of answers, and categorization of question styles.	Dimensions introduced to students indirectly include the dimensions of awareness, usage of evaluation and clinical ethics.	Semi-qualitative observation	Measured outcome	ChatGPT 3.5 interaction log and questionnaire	1 session or 15 min	Clinical case scenario	The lack of AI literacy training and structured prompt development in the curriculum for GenAI utilization
Fleischmann (2024)	74 Australian design students	Midjourney, Adobe Firefly, DALL-E, Bing Image Creator and ChatGPT	The prompt strategy used consists of prompt formulation exercises in a creative design studio and evaluation of the suitability of the results with the ethics of the work.	Dimensions introduced to students indirectly include the dimensions of awareness and usage.	Survey-based curriculum study	Self-reported	Survey	1 semester	Design studio project	The lack of a structured curriculum approach for GenAI integration and prompt design training in design studios.
Gong et al. (2025)	44 educational technology students	Kimi	The prompt strategy used is a four-stage progressive prompt, starting from situation exploration, program creation, model optimization and knowledge internalization, with attention to verification.	Dimensions introduced to students indirectly include the dimensions of awareness and evaluation.	Learning analytics	Measured outcome	Test and log HCI interactions	130 min	Python programming	The lack of comprehensive analysis on the effectiveness of general prompts and advanced prompts and their relationship to human-computer interaction patterns

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Table 3 (*continued*)

Author (Year)	Population	Types of GenAI	Prompt Strategy	AI literacy introduced to students	Method	Type of Evidence	Instrument	Duration	Task Type	Research Gap
										and students' computational reasoning.
Huang et al. (2024)	First-year students in the USA	ChatGPT and Gemini	The prompt strategy used includes copying, editing, and detailing the prompt, as well as requesting further explanation for verification.	Dimensions introduced to students indirectly include the dimensions of awareness, usage, evaluation and ethics.	Case Study	Mixed	ChatGPT 4.0 tests, surveys, and interaction logs	8 weeks	Formative assessment	Limited practical guidance and structured evidence for the use of GenAI in formative assessment in higher education
Hwang et al. (2025)	10 graduate students in English as a foreign language at level C2	ChatGPT	The prompt strategy used emphasizes response control by setting the text style, tone and length, contextualization, language selection, sentence and format setting, and compiling a list of output requirements that GenAI must meet.	Dimensions introduced to students indirectly include the dimensions of awareness, usage and evaluation.	Mix method	Mixed	ChatGPT interaction log	8 weeks	Academic writing assignments	The need for a validated prompt design strategy taxonomy and cross-task testing of language and broader populations.
Knoth et al. (2024)	45 German students	ChatGPT	The prompt strategy used emphasizes precision prompts with context and constraints, iteration, verification and prompt quality assessment.	Dimensions introduced to students indirectly include the dimensions of usage, evaluation and academic ethics.	Mixed method	Mixed	Integrative complexity score, prompt quality score, and AI literacy questionnaire	30 min	Travel plans and scientific project planning	There is still little research linking students AI literacy with prompt design skills and their impact on the quality of large language model output.
Partridge et al. (2024)	Students in the English language preparation program	ChatGPT	The prompt strategy used consists of effective prompt writing exercises accompanied by output evaluation.	Dimensions introduced to students indirectly include the dimensions of awareness, usage, evaluation, and academic ethics.	Cases study	Self-reported	Written reflection survey	1 semester	Academic writing assignments	The limited scope of the study, which was confined to a single class, meant that it did not examine the causal impact of AI literacy training and prompt design on learning outcomes and required a more structured empirical design.
Picardal (2025)	85 Filipino science teacher candidates	ChatGPT, Perplexity AI, Gemini	The prompt strategy used takes the form of structured intelligent instructions with clear roles, contexts and steps, as evidenced by verification.	Dimensions introduced to students indirectly include the dimensions of awareness, usage, evaluation and academic ethics.	Descriptive qualitative	Self-reported	AI interaction log	1 semester	Academic writing assignments in science education	There is a need for validated taxonomies and metrics to objectively assess smart prompt strategies, along with cross-course replication so that descriptive qualitative findings become more generally applicable.
Rovira-Esteva (2025)	32 first-year Spanish students	ChatGPT	The prompt strategy used is targeted use with an emphasis on ethics and responsibility	Dimensions introduced to students indirectly include the dimensions of awareness, usage and evaluation and ethics	Qualitative exploratory study	Self-reported	ChatGPT interaction log	1 semester	Contemporary China research assignment	The limited observations in uncontrolled situations with limited samples require standardized quantitative evaluation and AI literacy training interventions so that the risks and benefits of GenAI can be assessed more reliably

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Table 3 (*continued*)

Author (Year)	Population	Types of GenAI	Prompt Strategy	AI literacy introduced to students	Method	Type of Evidence	Instrument	Duration	Task Type	Research Gap
Sawalha et al. (2024)	54 embedded systems students from the United Arab Emirates	ChatGPT	The prompt strategy used maps the characteristics and quality of the prompt and tests its relationship with output accuracy.	Dimensions introduced to students indirectly include the dimensions of usage, evaluation and ethics of GenAI use.	Mix method	Measured outcome	ChatGPT test and interaction logs	1 semester	Embedded systems and programming tasks	Weak generalization due to a single domain and a single tool, requiring cross-disciplinary comparisons and controlled testing of the relationship between prompt design strategies and response accuracy and learning performance.
Quince and Nikolic (2025)	Australian engineering students	ChatGPT	The prompt strategy used requires specific and contextual prompts as well as gradual evaluation of bias and quality.	Dimensions introduced to students indirectly include the dimensions of usage and evaluation.	Phased assessment design	Mixed	Phased assessment	1 semester	Phased assignment of AI usage in engineering	The need for curriculum reinforcement and structured assessment that can balance understanding of the benefits and challenges of using artificial intelligence in higher education.

used *mixed* evidence, 5 relied on *self-reported* evidence, and only 3 used *measured outcomes*. These findings indicate that most of the available evidence is based on student perceptions, either independently or in combination with objective measurements. In addition, 4 of the 15 studies did not explicitly report the sample size in the available population description. The duration of the studies varied from a single session to a full semester, with the majority of studies lasting one semester. The types of evidence referred to are as follows: *measured outcome* refers to objective measurement instruments such as test scores, log interaction analysis, and assessment rubrics; *self-reported* refers to students' perceptions or subjective reports through surveys, interviews, and reflective journals; while *mixed* is a combination of the two. The distribution of these methodologies should be interpreted as an initial indication of existing research trends because the limited size of the corpus does not allow for definitive conclusions to be drawn.

4.4. Types of tasks in the use of prompt engineering

Findings regarding the types of tasks that are relevant and frequently use *prompt engineering* have been summarized in Table 3. Academic writing tasks were the most common type of task in four studies. This was followed by programming tasks, which appeared in three studies, and formative assessment tasks, which appeared in two studies. Other task types include general coursework, Spanish language assignments, clinical case scenarios, design studio projects, and contemporary Chinese studies assignments. Academic writing and programming tasks are the most common users of prompt engineering because they excel at guiding GenAI to generate text and code, thereby significantly enhancing the efficiency of these two task types (Black & Tomlinson, 2025). In academic writing assignments, user-inputted prompt engineering can guide GenAI to support the process of brainstorming ideas, outlining, drafting, and editing (Johnston et al., 2025). Similarly, in programming tasks, prompt engineering is positioned to efficiently guide GenAI in generating program code, explaining program logic, and correcting program code (Bringula, 2024). The distribution pattern of these task types should be viewed as preliminary findings influenced by the limited number of studies included, so further generalization requires the accumulation of additional empirical evidence.

4.5. Research gaps in prompt engineering research

Findings regarding research gaps have been summarized in Table 3. The research gap that has not been filled in this study is the lack of empirical evidence regarding the effectiveness of *prompt engineering* compared to direct teaching interactions by lecturers (Andewi et al., 2025). In addition, there have been no long-term studies tracing the evolution of students' *prompt* writing strategies over time, including the development of critical evaluation and ethical reflection skills in the use of AI models (Aruleba et al., 2025). Methodologically, many previous studies have limitations, such as a lack of research in controlled learning situations and small sample sizes (Partridge et al., 2024; Rovira-Esteva, 2025). Furthermore, there is no comprehensive analysis of the differences in effectiveness between general prompts and advanced prompts and their impact on human-computer interaction patterns (Gong et al., 2025).

In addition to research design gaps, there are also gaps related to the scope and generalizability of findings. Understanding of the factors that influence the adoption of GenAI in higher education is also limited, so the existing digital competency framework needs to be updated to include new skills, such as prompt engineering and GenAI output evaluation (Caner-Yildirim, 2025). Previous research has minimally explored prompt engineering in languages other than English. It is important to know whether prompt composition techniques in non-English languages can improve the quality of GenAI responses (Carrasco-Sáez et al., 2025). At the curriculum level, AI literacy and prompt engineering training have not been systematically integrated, leaving educators lacking evidence-based practical guidance on utilizing GenAI in formative assessment (Desseauve et al., 2024). Furthermore, many findings are specific to a single domain or type of AI tool, limiting their generalizability (Sawalha et al., 2024).

Regarding instruments and measurement, this review also found significant gaps. To bridge this gap, it is necessary to develop a validated prompt engineering taxonomy along with objective evaluation metrics so that the use of prompt engineering can be measured consistently (Hwang et al., 2025). The presence of a standard taxonomy framework and evaluation metrics also enables cross-testing of task types, courses, and student populations so that research findings can be more easily generalized (Picardal, 2025). This condition indicates the need for cross-disciplinary research with controlled experimental designs and standardized quantitative evaluations to more reliably assess the relationship between *prompt* composition strategies, AI response accuracy, and student learning performance. In addition, AI literacy training interventions need to be systematically tested so that the benefits and risks of GenAI implementation in learning can be evaluated more objectively (Quince & Nikolic, 2025). Overall, these research gaps underscore the urgent need for controlled and longitudinal comparisons between prompt engineering and direct instruction, the development of standardized assessment instruments, cross-disciplinary and cross-linguistic replication, and the curricular integration of AI literacy and ethical governance, so that causal evidence of learning impacts can go beyond descriptive findings.

4.6. *Thematic analysis*

All empirical studies were analyzed thematically, then the results of 15 empirical studies were synthesized and categorized, leading to six main titles related to *prompt engineering* to bridge student AI literacy in higher education. These six themes should be understood as initial patterns that emerge from the available evidence and have the potential to mature as empirical findings in this field increase.

4.6.1. *Prompt engineering in higher education learning*

Prompt engineering in higher education learning is intended to produce GenAI outputs that are appropriate for the contextual goals of education (Lee & Palmer, 2025), as demonstrated by empirical studies confirming that prompting techniques influence the quality of GenAI responses (Carrasco-Sáez et al., 2025) and can support student proficiency (Andewi et al., 2025). Based on preliminary evidence from the reviewed studies, good prompt engineering usage begins with role assignment, formulation of the desired context, setting boundaries, followed by requesting explanations and verification or by writing down sources (Aruleba et al., 2025). Role definition, context formulation, and scope limitations are reported to be associated with increased answer accuracy on tasks that require precision (Carrasco-Sáez et al., 2025). The quality of GenAI output tends to be more relevant when students add constraints that are not already listed, clarify the desired output format, or request alternative perspectives.

(Aruleba et al., 2025; Picardal, 2025). Requests for GenAI to explain answers step by step and perform verification aim to provide students with systematic output notes that can be matched with lecture theories and verified against reliable sources (Aruleba et al., 2025). The improvement in output results is not due to an increase in GenAI's capabilities to become smarter, but is associated with the development of students' skills in directing conversations better through prompt engineering (Yan et al., 2025). Interacting with GenAI can produce relevant and consistent outputs when following templates and rubrics to measure the quality of prompts used as guidelines in the prompt engineering process (Knoth et al., 2024). The prompt engineering template includes a structure consisting of role assignment, context formulation, boundary determination, output format, explanation requests, and verification.

4.6.2. *The effect of prompt engineering on AI literacy in learning*

The proper application of *prompt engineering* in GenAI-integrated learning aims to provide students with adequate AI literacy, one of which is to be able to critically assess the output generated by GenAI before use (Knoth et al., 2024). Preliminary evidence indicates that the application of prompt engineering when interacting with GenAI in the classroom can facilitate students' understanding of the dimensions of AI literacy, because the structure of prompt engineering includes elements that indirectly encourage students to pay attention to the dimensions of AI literacy. The first dimension is awareness, which is evident when students understand the limitations of using GenAI (Carrasco-Sáez et al., 2025; Knoth et al., 2024). These limitations are intended to make students aware of the capabilities, risks, and appropriate timing of GenAI use in higher education (Knoth et al., 2024). Second, *prompt engineering* skills, which begin with role assignment, context formulation, and iterative conversation management, including documenting *prompt* changes, indirectly make students pay attention to the dimensions of GenAI use (Knoth et al., 2024). Third, there is the dimension of evaluating and verifying the output of GenAI, which is evident when students enter a prompt requesting an explanation and perform verification, including checking facts and data, comparing with scientific readings, and providing reasons for accepting, changing, or rejecting the output from GenAI (Aruleba et al., 2025; Huang et al., 2024; Knoth et al., 2024; Picardal, 2025). Fourth, ethics and responsibility, which include academic integrity, proper attribution, awareness of bias, and social impact (Aruleba et al., 2025). Students who understand AI literacy can critically assess the quality of GenAI output without relying on and simply accepting the results (Picardal, 2025). However, the available evidence still needs to be confirmed through more rigorous experimental studies. Despite these limitations, prompt engineering can still be a practical bridge for introducing AI literacy in learning. AI literacy does not replace mastery of the material, but it makes learning with GenAI safer and more productive (Caner-Yıldırım, 2025; Knoth et al., 2024; Partridge et al., 2024).

4.6.3. *Integrated learning design prompt engineering in higher education*

Preliminary findings indicate that learning design influences the use of *prompt engineering* and AI literacy, enabling students to become accustomed to using them when interacting with GenAI (Andewi et al., 2025; Knoth et al., 2024). Students can become accustomed to using prompt engineering when interacting with GenAI through guided learning (Hwang et al., 2025). Guided learning appears to make the process transparent, such as when teachers clearly teach prompt engineering, verify GenAI outputs, and then invite reflection so that students explain the decisions in each of their interactions with GenAI. The guided learning design includes process-based assessment that can align learning incentives with learning objectives (Huang et al., 2024). The process assessed from students is the prompt engineering found in the interaction log with GenAI, including the reasons for revisions and evidence of verification of GenAI outputs, which are then assessed by the instructor (Knoth et al., 2024). This approach has been reported to motivate students to document the reasons behind each decision and critically evaluate GenAI outputs (Fleischmann, 2024). However, these findings are still based on a small number of studies, so their effectiveness needs to be tested across a wider range of disciplines, educational levels, and student characteristics. Despite these limitations, guided learning design still shows potential for making learning with GenAI transparent and monitorable by instructors through the use of prompt engineering (Fleischmann, 2024; Huang et al., 2024; Knoth et al., 2024).

4.6.4. *Student performance and learning outcomes*

The application of learning design is reported to facilitate students in using *prompt engineering* when interacting with GenAI (Andewi et al., 2025). Preliminary evidence from the reviewed studies indicates that students' understanding of lecture material can be aided, ultimately leading to improved learning outcomes (Picardal, 2025). Research reports that the integration of prompt engineering for GenAI in writing courses is associated with an increase in students' vocabulary and grammar mastery (Andewi et al., 2025). Furthermore, in the context of problem-solving tasks, the accuracy of answers tends to be higher when the prompt input into GenAI requires structured reasoning and specifies relevant constraints (Sawalha et al., 2024). However, these findings are still preliminary given the limited number of studies, and improvements do not occur in all domains. In clinical environments or other sensitive contexts, accuracy remains vulnerable without strict verification and governance (Desseauve et al., 2024). The benefits of learning integrated with GenAI are highly dependent on the suitability of task demands, prompt engineering, AI literacy, and learning design (Andewi et al., 2025; Knoth et al., 2024).

4.6.5. Student attitudes and behavior towards prompt engineering

Students' actual intention to use GenAI is influenced by factors such as habit, convenience, performance benefit expectations, and support in the form of technology access and training (Caner-Yıldırım, 2025). However, in practice, many students use GenAI freely without considering AI literacy and often rely only on general prompts (Knoth et al., 2024). This unfocused approach to usage has the potential to cause negative impacts, such as students accepting incorrect GenAI outputs that may be biased without first verifying them (Kofinas et al., 2025). Students' attitudes toward the use of GenAI in an academic context show diversity. Some students show enthusiasm because this technology can increase the efficiency of completing assignments, but there are also many who doubt the accuracy and authenticity of the results and are concerned about the potential for academic cheating (Fleischmann, 2024). Although these findings come from diverse contexts and populations, their generalization needs to be considered carefully; there is a common point that strengthening AI literacy, especially through prompt engineering, can help students produce relevant GenAI outputs.

4.6.6. Risks and limitations of generic prompts in learning

The use of generic prompts in GenAI in higher education carries a number of risks that have been identified by various studies. First, GenAI is prone to producing inaccurate outputs due to hallucinations, i.e., fictional information presented as accurate, which has the potential to mislead students' understanding (Alkaissi & McFarlane, 2023; Ji et al., 2023). Second, the use of generic prompts often triggers inappropriate or inaccurate responses, requiring students to spend a long time using GenAI to assist with their assignments (Huang et al., 2024). Third, GenAI can cause students to become overly dependent on instant AI feedback, which encourages superficial learning engagement and weakens their ability to analyze independently (Ododo et al., 2024; Zhai, 2024). Academic ethical challenges also arise, such as the risk of plagiarism and academic integrity violations, especially if students accept AI answers without critical verification (Barrett & Pack, 2023; Chan & Hu, 2023). To mitigate the above risks, the literature emphasizes the importance of improving AI literacy and understanding of prompt engineering among educators and students in higher education (Knoth et al., 2024; Partridge et al., 2024). The use of *prompt engineering* is reported to be associated with increased accuracy and contextuality of GenAI responses (Hwang et al., 2025; Sawalha et al., 2024), while adequate AI literacy is expected to help users understand the risks and limitations, such as bias and ethical issues associated with AI use (Aruleba et al., 2025; Knoth et al., 2024; Yan et al., 2025). In turn, this allows AI to be used as a supporting tool that enriches the learning process without replacing the essence of student engagement and critical thinking in higher education (Gong et al., 2025; Huang et al., 2024).

5. Discussion

This study aims to map the relationship between the prompt engineering used by students and the dimensions of AI literacy. Available evidence indicates that the use of GenAI without adequate prompt engineering tends to produce irrelevant outputs and potentially has negative impacts on students who do not yet understand AI literacy (Güner & Er, 2025). Several previous reviews have mapped different aspects of GenAI use in higher education, including changes in assessment at the student, instructor, and institutional levels (Xia et al., 2024), the use of GenAI in language learning (Wang et al., 2025) and the utilization of ChatGPT for teaching and learning (Liu et al., 2025). From a conceptual perspective**,** Lai and Tu (2024) discuss the role of GenAI in the learning environment, while Tu and Lu (2025) identify that studies on the cognitive and behavioral characteristics of students during GenAI-based learning are still lacking. These reviews differ substantially from this study, which specifically maps empirical research on prompt engineering as a bridge for AI literacy among college students, a topic where empirical research is still limited and its findings are scattered.

Before detailing the findings, it is important to emphasize that all interpretations in this section are based on the corpus of empirical studies that meet the inclusion criteria. The limitations in the number and heterogeneity of study designs constrain the power of inference and generalization, so the following findings should be understood as preliminary evidence that provides direction for further research. Analysis of the definitions of prompt engineering in the literature reveals two complementary approaches: explicit and implicit. This distinction arises from differences in the objectives and methods across studies. Explicit definitions formulate prompt engineering as the process of designing, formulating, and optimizing prompts so that GenAI produces the desired, accurate, and contextual output. When researchers target trainable and measurable learning, they require a clear operational definition, so prompt engineering is articulated as a complete technical procedure, including prompt characteristics, monitoring of interaction logs, and evaluation rubrics to assess the quality and bias of the output (Carrasco-Sáez et al., 2025; Knoth et al., 2024). This aligns with the goal of explicit learning, which is the process of acquiring conscious, verbalizable knowledge through formal instruction (Ellis, 2009; Kim & Godfroid, 2023). Therefore, training-or task-analysis-designed studies tend to use an explicit framework and report tentative associations between prompt quality and task completion accuracy and learning outcomes.

Conversely, other studies implicitly define prompt engineering as part of GenAI usage competencies within an AI literacy framework integrated into the educational curriculum without being defined separately. When prompt engineering is incorporated into broader learning objectives and ethical guidelines, its mastery occurs through an unconscious process, but this knowledge is reflected in automatic performance (Rebuschat & Williams, 2012; Weinberger & Green, 2022). The difference between these two approaches reflects that this field is still in its early stages of development, as each study defines the concept differently due to varying objectives and methods. These differences in definition have practical implications in that it is difficult to compare or replicate results between studies. Studies with explicit definitions, such as Knoth et al. (2024) and Carrasco-Sáez et al. (2025), do produce more measurable data through rubrics and interaction logs, but they fail to capture how *prompts* are actually used naturally in the classroom. Conversely, studies with implicit definitions, such as Fleischmann (2024), provide a more contextual picture, but are difficult to replicate because the conceptual boundaries are not operationally defined.

Regardless of these definitional differences, prompt engineering has been applied across disciplines and has been reported to have the potential to support learning processes in various higher education contexts (Federiakin et al., 2024). Its application is most commonly found in English language learning, as the text-based nature of the tasks allows rhetorical goals, style, and audience to be clearly and purposefully formulated and then organized into specific prompts for GenAI. This is reflected in an experimental study reporting that directed interaction with ChatGPT in argumentative essay writing is associated with increased motivation among second language learners (Zare et al., 2025). Additionally, *prompt engineering* is also widely applied in computer science education in higher education because the nature of programming tasks is highly dependent on detailed *prompts* to generate correct code, explain logic, and correct errors (Bringula, 2024). A quasi-experimental study in a programming course reported that the group that integrated ChatGPT using clear prompts showed more structured work behavior, indications of improved programming task performance, and more positive learning perceptions compared to the group without clear prompt integration (Sun et al., 2024).

The identified *prompt* strategy to bridge student AI literacy follows four stages. First, role definition aims to minimize context ambiguity, set boundaries for knowledge and evaluation style, and reduce response variation between sessions, thereby facilitating more structured iteration and reflection. This is supported by an experimental study evaluating "role-playing prompts" that reported improved model performance on domain-based question-answering tasks when roles were defined in advance (Chen et al., 2025). Second, the formulation of specific contexts and tasks to reduce ambiguity and response variation, while maintaining consistency in style and depth of discussion (Hwang et al., 2025; Partridge et al., 2024). This is in line with findings that studies on language learning plan creation show that the level of specificity and context in instructions is reported to be associated with the quality and stability of GenAI output (Dornburg & Davin, 2024). Third, setting boundaries to direct GenAI outputs in accordance with predetermined contextual and task objectives, such as work steps, scope, and assessment criteria, so that responses are relevant and easily evaluated against learning rubrics (Hwang et al., 2025; Knoth et al., 2024), with initial indications of the effectiveness of ChatGPT-based scaffolding when the prompt is narrowed by structured steps (Liao et al., 2024). Fourth, requests for explanation and verification are a critical stage often found in the reviewed studies (Andewi et al., 2025; Aruleba et al., 2025), given that students reportedly rarely request verification of the credibility of ChatGPT outputs independently, which is only about 1.6 percent of 308 requests in IELTS writing tasks. so that *prompt* strategies that explicitly request clarification and justification need to be designed to encourage students to assess the reliability of the output before using it (Rapp et al., 2025; Zhan & Yan, 2025). The fourth stage of this sequence was identified from patterns that emerged in the reviewed studies and needs to be positioned as a tentative framework that requires further empirical validation.

Beyond the sequence of strategies, the review findings also indicate a relationship between the application of prompt strategies and students' AI literacy dimensions. Knoth et al. (2024) report that the application of prompt engineering can also make students pay attention to AI literacy dimensions. The study reports that aspects of AI literacy correlate with the quality of prompt engineering practices and model adaptation in higher education environments, tentatively indicating that when students design prompts in a more focused manner, they activate AI literacy dimensions, particularly in the areas of operational use and evaluation (Knoth et al., 2024). AI literacy has four dimensions, namely the awareness dimension, the usage dimension, the evaluation dimension, and the ethics dimension (Wang et al., 2023). The findings indicate that attention to the ethics dimension in higher education appears to be the lowest among the other AI literacy dimensions based on the studies reviewed. This pattern needs to be interpreted with caution, given the limited and heterogeneous corpus, but its consistency is strong enough to warrant further explanation of the underlying factors.

The low attention to the ethical dimension can be explained by three interrelated factors. First, *prompt engineering* research in higher education is still dominated by a technical effectiveness orientation, where the success of *prompt engineering* interventions is measured through output accuracy, task completion, and academic performance (Knoth et al., 2024; Lee & Palmer, 2025), so that the ethical dimension has not become a major variable that is systematically examined in research design. Second, although general AI ethical principles have been formulated, such as benevolence, non-harm, fairness, transparency, and autonomy (Floridi et al., 2018), an ethical evaluation framework specifically designed to assess prompt engineering practices in the context of learning is not yet available in the literature. The absence of this specific evaluation framework makes it difficult for researchers to operationalize ethical dimensions as variables that can be measured and compared across studies. Third, the nature of the field, which is still in its nascent phase, encourages researchers to prioritize proving the pedagogical effectiveness of prompt engineering before examining its implications in depth. This pattern of prioritization is common in developing fields and is consistent with the findings of Tu and Lu (2025), who, through keyword co-occurrence analysis, identified that GenAI research in the education sector is still dominated by studies on perception and effectiveness, while ethical dimensions and student behavior have received less attention. Therefore, the low level of attention to the ethical dimension does not indicate the irrelevance of this dimension, but rather reflects the maturity stage of the field, which has not yet made ethics a main focus of study.

Regardless of the factors explaining this lack of attention, the neglect of the ethical dimension still has consequences. The use of GenAI in higher education without considering the ethical dimension can increase the risk of academic integrity violations, including academic cheating and fraud (Rudolph et al., 2023). This finding is reinforced by the perspective of an expert in the field of educational computing, Hwang (in Allison et al., 2025), who explicitly recommends investigating ethical issues as a priority research focus, emphasizing that the use of GenAI in education not only opens up opportunities to improve learning performance but can also raise ethical issues, including the practice of students using GenAI answers as their own work. More broadly, Allison et al. (2025) place ethics, social sustainability, and well-being as one of the four main themes that require ongoing research attention in educational computing (Allison et al., 2025).

The urgency of this ethical dimension has actually been identified since the inception of the AIED field. Hwang et al. (2020) in the inaugural editorial of CAEAI established the development of ethical principles and practices as one of ten priority research topics because the application of AI in education has the potential to raise ethical issues, such as digital hegemony and the digital divide. Based on this urgency, prompt engineering can play a role in operationalizing AI ethical principles into concrete learning practices in higher education. These principles refer to the ethical framework of Floridi et al. (2018), which includes five principles: benevolence, non-maleficence, justice and equality, transparency and accountability, and autonomy. Structurally, prompt engineering supports non-harmful practices because its structure includes components that facilitate users in verifying the expected output of GenAI, which can foster a sense of responsibility and prevent academic fraud and cheating (Cotton et al., 2023). Prompt engineering, in its structure, already supports the four dimensions of AI literacy, namely the awareness dimension, which is evident when students understand the capabilities, risks, and time required in using GenAI (Hwang et al., 2025; Knoth et al., 2024), and the usage dimension, which is evident when writing role assignments, formulating contexts, and managing conversations iteratively (Andewi et al., 2025; Hwang et al., 2025; Knoth et al., 2024), the evaluation dimension, which is evident when students request explanations and verify GenAI outputs (Aruleba et al., 2025; Huang et al., 2024; Knoth et al., 2024; Picardal, 2025) and the ethical dimension that includes academic integrity by critically considering the use of outputs to foster a sense of responsibility while preventing academic cheating (Aruleba et al., 2025; Fleischmann, 2024).

It should be noted that the support of prompt engineering for these evaluation and ethical dimensions is not automatic and may even be counterproductive without adequate pedagogical supervision. The first risk is automation bias, which occurs when students assume that well-structured prompts automatically produce accurate outputs, thereby neglecting critical verification. Early indications of this phenomenon are evident in the findings of Zhan and Yan (2025), who reported that only about 1.6% of 308 student requests asked for verification of the credibility of ChatGPT's suggestions, suggesting that overconfidence in GenAI outputs may already exist even without structured prompts and tends to strengthen when students feel they have used the "right" prompts. The second risk is *cognitive offloading,* where students delegate analytical thinking processes to GenAI through structured prompts, rather than using GenAI as a tool to deepen their own understanding. This concern is consistent with findings that over-reliance on instant GenAI feedback can encourage superficial learning engagement and weaken independent analytical skills (Fan et al., 2025), and aligns with Hwang's (in Allison et al., 2025) warning that *educational computing* research tends to focus too much on easily measurable outcome metrics and neglects students' thinking processes. The third risk is *dependency*, which occurs when the ease of producing high-quality output through *prompt engineering* reduces students' motivation to complete tasks independently. This risk is in line with Hwang's warning about *metacognitive laziness* (Allison et al., 2025; Fan et al., 2025), a condition in which students stop trying to complete tasks independently because they are too dependent on GenAI assistance. These three risks are mutually reinforcing because automation bias reduces critical awareness, cognitive offloading reduces cognitive investment, and dependency erodes independent motivation, thus cumulatively potentially eroding the dimensions of evaluation and ethics in the AI literacy framework. Therefore, the effectiveness of prompt engineering in supporting the ethical dimension is highly dependent on pedagogical interventions accompanying its use, ensuring that students are not merely skilled at composing *prompts* but also remain critical, reflective, and responsible for the outputs generated.

Analyses of findings on the *prompt engineering* method in the literature mostly use qualitative and mixed methods (Lee & Palmer, 2025). This dominance can be explained by the exploratory nature of the field and its highly context-dependent application, where the success of GenAI outputs is still subjective and varies according to the task or discipline. The lack of standardized prompt skill assessment instruments has encouraged researchers to rely on qualitative approaches to capture processes and meanings, then combine them with quantitative measures as triangulation. The most widely studied GenAI is ChatGPT, reflecting its rapid adoption since its public release in November 2022, with over 100 million active users (Wang, Chen, Wang, et al., 2024). While this approach is appropriate for the stage of development of the field, this dominance indicates that causal evidence regarding the effectiveness of prompt engineering in bridging AI literacy is still very limited. In the absence of randomized controlled trials (RCTs) that rigorously compare structured prompt strategies with unguided GenAI interactions, claims of effectiveness in the current literature should be understood as tentative associative relationships rather than confirmed causal relationships.

The most common type of task in the literature is academic writing, where prompt engineering is reported to be associated with potential benefits in idea development, content organization, and grammatical accuracy through metalinguistic explanations (Mahapatra, 2024). Prompt engineering is also applied to student programming tasks, but its formulation requires specific, step-by-step prompts and the ability to break down problems into appropriate questions, which is the core of prompt engineering (Jing et al., 2024). In programming tasks, prompt engineering is positioned to efficiently guide GenAI in generating program code, explaining program logic, and correcting program code (Bringula, 2024). In formative assessment tasks, prompt engineering is used by defining the role of the model, providing task context, embedding criteria, limiting output formats, and requesting actionable reasons and revision suggestions (Wang, Chen, Wang, et al., 2024). The concentration of studies on academic writing and programming tasks is understandable because these two types of tasks have characteristics that are structurally most compatible with prompt-based interactions, namely: both are text-based, have relatively definable success criteria, such as grammatical accuracy, code correctness, or structural appropriateness, and allow for repeated prompt iterations where students can refine the prompt based on evaluation of previous outputs (Mutanga et al., 2025; Qian, 2025; Schulhoff et al., 2024). However, the dominance of these two task types raises important questions about whether the effectiveness of prompt engineering can be generalized to less structured tasks with more ambiguous success criteria, such as creative design tasks (Creely & Blannin, 2025; Ivcevic & Grandinetti, 2024), open-ended problem solving (Boussioux et al., 2024), ethical reasoning, or collaborative project-based tasks involving multi-perspective negotiation (Doshi & Hauser, 2024). The limitations of the scope of these task types need to be considered when interpreting the practical implications of the review findings, as the observed effectiveness patterns of *prompt engineering* in structured tasks may not necessarily be directly transferable to fundamentally different task contexts (Federiakin et al., 2024; Lee & Palmer, 2025).

An analysis of research gaps in prompt engineering research in relation to AI literacy identifies a lack of empirical evidence regarding the potential effectiveness of prompt engineering compared to direct teaching interactions by lecturers (Andewi et al., 2025). This pattern is understandable given that most existing research still focuses on evaluating the quality of GenAI output and its ability to provide feedback (Jacobsen et al., 2025). Furthermore, there is a research gap in that there are no long-term studies tracing the development of students' *prompt* writing strategies over time (Aruleba et al., 2025). This pattern can be explained by several factors, as current research is dominated by snapshot studies on specific tasks, validation in real classrooms is still limited, replication is hampered by the inaccuracy of GenAI outputs, and ethical and privacy issues hinder long-term tracking (Chen et al., 2024). Furthermore, there is no comprehensive analysis of the differences in effectiveness between general prompts and advanced prompts and their impact on user interaction patterns with GenAI (Gong et al., 2025). The lack of comprehensive analysis is understandable given that existing research is still focused on model performance on benchmarks (Jeon & Kim, 2025).

In addition to empirical gaps, this review also identifies gaps related to the limited scope of research. There is still a lack of research exploring prompt engineering in languages other than English, even though it is important to know whether prompt engineering techniques in non-English languages can improve the quality of GenAI responses (Carrasco-Sáez et al., 2025). Preliminary findings indicate that the effectiveness of prompt strategies depends on the language used, requiring non-English empirical research to test whether prompt designs using non-English languages improve the quality of GenAI responses (Xuan et al., 2025). At the curriculum level, AI literacy and prompt design training have not been systematically integrated, resulting in educators lacking evidence-based practical guidance on utilizing GenAI in formative assessment. Consequently, the use of GenAI in learning evaluation remains sporadic and unstructured (Desseauve et al., 2024). Although lecturers are beginning to understand the potential of GenAI in learning, its application for assessment is still limited and unstructured, indicating the need for clear policies and curricular training to optimize AI literacy and *prompt* writing skills in higher education (Lyu et al., 2025).

In terms of methodology, a number of research design weaknesses were also identified in the studies reviewed. Methodologically, many previous studies had limitations, such as a lack of research in controlled learning situations and small sample sizes (Partridge et al., 2024; Rovira-Esteva, 2025). In addition, many findings were specific to only one domain or one type of AI tool, thus limiting their generalizability (Sawalha et al., 2024). This is evident in a study comparing the assessments of chatbots, lecturers, and peers in a single course at a single institution with 76 students, without controlled learning conditions or randomization, specific to a particular tool, and not longitudinal, thus limiting its generalizability. The author also emphasizes these limitations and recommends larger samples, more diverse contexts, and evaluation of the impact of feedback on the quality of students' final results in further research (Usher, 2025).

These research gaps should be understood as preliminary indications based on currently available evidence and may change as more empirical evidence becomes available. The persistence of these research gaps is understandable given that the field of prompt engineering in higher education is still in its early stages of development, with terminology that is not yet uniform and conceptual understanding that is still fragmented (Schulhoff et al., 2024), so that initial mapping efforts through systematic reviews have only begun in recent years (Lee & Palmer, 2025; Qian, 2025). In terms of measurement, Federiakin et al. (2024) show that the current 21st-century skills framework does not adequately cover prompt engineering to enable valid assessment, while in the broader context of AI literacy, Lintner (2024) found that although a number of AI literacy scales show good structural validity and internal consistency, most have not been adequately tested in terms of content validity, reliability, construct validity, and cross-cultural validity. This condition helps explain why the available empirical studies are generally still short-term and self-report data-based, as emphasized by several researchers who call for the need for longitudinal designs to investigate the long-term impact of GenAI devices on learning (Lee, S. et al., 2025). In addition, the dominance of using one platform, namely ChatGPT, also narrows the diversity of findings. Lee, S. et al. (2025) note that almost all of the studies reviewed used ChatGPT as their primary tool, a trend that is in line with students' perceptions of the ease of access and use of GenAI tools (Sousa & Cardoso, 2025), but this makes it difficult to generalize the research results to other GenAI platforms such as Gemini, Claude, or Copilot.

These gaps also align with broader concerns within the educational computing community. Huang (in Allison et al., 2025) emphasizes that the current research orientation needs to shift from simply measuring AI-enabled outcomes to understanding how AI reshapes thinking processes. This perspective reinforces the review's finding that studies are needed that not only measure the effectiveness of prompt outputs but also examine how prompt engineering shapes students' cognitive processes, including critical thinking, problem solving, and metacognitive reflection skills that are central to AI literacy. This gap is also confirmed by Tu and Lu (2025), who, through *keyword co-occurrence* analysis, found that GenAI research in education still lacks analysis of students' cognitive and behavioral characteristics during the learning process, as well as a lack of studies on GenAI as an evaluation and feedback tool. Overall, these gaps reflect the general characteristics of a research field that is still in the foundation-building phase (Lee & Palmer, 2025; Qian, 2025), and underscores the urgent need for controlled and longitudinal comparisons between *prompt* strategies and direct instruction, the development of standardized assessment instruments, cross-disciplinary and cross-linguistic replication, and the curricular integration of AI literacy alongside ethical governance to ensure that causal evidence of learning impacts goes beyond descriptive findings.

6. Implications

This scoping review provides valuable insights for higher education stakeholders in promoting the use of prompt engineering when interacting with GenAI in higher education. The use of prompt engineering is intended to bridge students' AI literacy in using GenAI responsibly, minimize the risk of inaccurate GenAI outputs, and foster a habit of evaluation and reflection among students (Knoth et al., 2024). Theoretically, this review positions the link between prompt engineering and AI literacy as an evolving and dynamic concept. Practically, the identified implications can be understood within the framework of AI's role in education formulated by Hwang et al. (2020), namely AI as an intelligent tutor, intelligent tutee, intelligent learning tool/partner, and policy-making advisor. Through appropriate prompts, GenAI can be directed to function as a tutor that provides personalized feedback, a learning tool that aids in knowledge exploration, or a tutee that is taught by students, thereby deepening their own understanding. The implications for each stakeholder are outlined as follows.

For policymakers, the findings of this review indicate that the ethical dimension is the least considered dimension of AI literacy in the studies reviewed (53%), while Zhan and Yan (2025) report that only about 1.6% of 308 student requests asked for verification of the credibility of ChatGPT outputs. This low level of attention, compounded by the risks of automation bias and cognitive offloading, indicates that general GenAI usage policies in the form of prohibitions or permissions without operational guidelines are insufficient to protect students' academic integrity. Within the framework of Hwang et al. (2020), this recommendation is in line with GenAI's role as a policy-making advisor that requires ethical governance in order to function productively in the educational ecosystem. Operationally, each GenAI-based assignment should include: (1) a verification stage where students compare GenAI outputs with trusted sources and document their verification process, and (2) a written ethical reflection documenting considerations of academic integrity and potential bias in the adopted outputs. Institutional policies also require operational guidelines that include standard verification procedures, criteria for attributing GenAI outputs, and mechanisms for reporting academic integrity violations related to GenAI use. However, these policy recommendations are based on preliminary evidence from a still limited corpus, so their implementation needs to be adapted to each institutional context and evaluated continuously as empirical evidence in this field grows.

For lecturers, the findings of this review indicate that teaching prompt engineering needs to follow a scaffolding sequence derived from the identified prompt strategies, namely: (1) role assignment, (2) context and task formulation, (3) boundary setting, and (4) request for explanation and verification. This sequence provides a scaffolding framework that lecturers can use to design learning activities in stages, where students are not immediately asked to produce the final output, but are guided through each stage of prompt development progressively. Within the framework of Hwang et al. (2020), this approach activates the role of GenAI as an intelligent tutor that provides feedback at each stage of prompt development, as well as an intelligent learning tool that helps students explore knowledge in a structured manner. This review indicates that the assessment of prompt engineering activities needs to be process-based, not just based on the final output. Assessment based on interaction logs, reasons for revisions, and verification evidence has the potential to motivate students to document the reasons behind each decision and critically evaluate GenAI outputs (Fleischmann, 2024; Knoth et al., 2024).

For GenAI tool developers, the dominance of ChatGPT without built-in verification features, combined with low student self-verification (Zhan & Yan, 2025), indicates a design gap that needs to be addressed for educational contexts. Based on these findings, GenAI tools intended for educational contexts need to include at least three features derived from the review findings. *First*, an *auto-verification prompt* feature that automatically displays verification questions before students adopt the output, for example through questions such as “Have you verified this information with a reliable source?” or “Identify one potential weakness in this response.” This feature is based on the findings of low self-verification identified in the review. Second, a scaffolded prompt template guides users through a sequence of role assignment, context formulation, boundary setting, and verification, as per the sequence of prompt strategies identified in the findings of RQ2 of this review. This template allows GenAI to function as an *intelligent tutee* (Hwang et al., 2020), where the process of filling in each component of the template encourages students to articulate their understanding before receiving the output. *Third*, the *transparency indicator* is an indicator that shows the level of confidence or limitations of GenAI output to encourage critical evaluation by users, based on the findings of *automation bias* risks identified in the review.

For researchers, this review identifies a number of specific research gaps that require further investigation and generates a research agenda derived directly from the findings. First, the absence of randomized controlled trials (RCTs) comparing structured prompt engineering with

GenAI interactions without prompt engineering guidance indicates the need for experimental research measuring the extent to which structured prompt engineering strategies are more effective than general prompts in improving students' AI literacy evaluation dimensions. This research direction is important because all evidence in the reviewed corpus is still associative and requires causal confirmation. Second, the low attention to ethical dimensions and the absence of a specific ethical evaluation framework for prompt engineering practices indicate the need for research exploring how the explicit integration of ethical reflection in prompt engineering design affects students' awareness of automation bias and compliance with academic integrity principles. This research direction is in line with the recommendations of Allison et al. (2025), which place ethics as one of the priority themes in educational computing research. Third, the lack of prompt engineering studies in non-English languages suggests the need for research that tests the effectiveness of prompt engineering when formulated in non-English languages, including identifying the linguistic factors that mediate these differences. The preliminary findings of Xuan et al. (2025) also indicate that the effectiveness of prompt strategies may depend on the language used. Fourth, the absence of longitudinal studies in the reviewed corpus indicates the need for research that traces the development of prompt engineering and AI literacy skills over an academic year. The future research directions identified from this review are also in line with the research agenda collectively recommended by 14 educational computing experts. Allison et al. (2025) explicitly formulate research questions about how prompt engineering can be used as a pedagogical tool to develop students' computational thinking and problem-solving skills. These research questions confirm the relevance and urgency of further research on prompt engineering in higher education, particularly that which explores its relationship with dimensions of AI literacy.

7. Limitations

This scoping review has several limitations. First and foremost, the limited number of studies that met the inclusion criteria is a limitation that must be considered when analyzing all of the findings of this review. However, as emphasized by Arksey and O'Malley (2005), a scoping review aims to map the breadth of evidence available in a field, not to statistically calculate or weigh the quality of evidence. The PRISMA-ScR guidelines (Tricco et al., 2018) also do not set a minimum number of studies for a scoping review, so the size of the corpus is determined by the availability of empirical evidence in the field under review. This limited corpus restricts the ability to draw broad generalizations, perform meaningful statistical comparisons between subgroups, and ensure that the observed patterns are robust and replicable. Nevertheless, the limited size of the corpus confirms that this review is an initial mapping of an evolving academic landscape, where the small number of studies highlights the relevance of this review as a first step in synthesizing the available evidence to provide direction for further research. This is consistent with the findings of Tu and Lu (2025), which show that GenAI research in education is currently dominated by studies on student perceptions and literature reviews, while empirical studies on specific aspects, such as prompt engineering for AI literacy, are still very limited. In this context, this review is an initial mapping of the academic landscape. Second, not all studies included in this review are longitudinal studies. Only one study used a short-term longitudinal design with repeated measurements. The absence of long-term longitudinal studies limits our understanding of the long-term impact of using prompt engineering for context-appropriate GenAI output and for improving student AI literacy in higher education. Third, many of the studies reviewed used ChatGPT as a GenAI tool. This may limit the diversity of GenAI tools that can be used. The concentration on one type of GenAI tool, combined with the limited corpus size, further narrows the generalizability of the findings to more diverse GenAI usage contexts. Fourth, the exclusion of non-English literature may introduce potential bias. This may overlook valuable insights from non-English research. Fifth, this review only used three databases, namely Scopus, ERIC, and PubMed. Nevertheless, the three selected databases complement each other in covering literature from the fields of general education, educational technology, and health professional education. This limitation, as well as the language limitation in the fourth point, has the potential to narrow the scope of the literature covered and needs to be considered when interpreting the overall results of the review.

8. Future works

This scoping review provides a basis for supporting future primary research. Future research could expand the application of prompt engineering in higher education to fill the gaps identified in the previous discussion, particularly the lack of long-term studies that track the development of student prompt strategies over time and the application of AI literacy frameworks for the ethical use of GenAI in higher education. At the curriculum level, AI literacy frameworks and prompt engineering training have not been systematically integrated. Therefore, policymakers and faculty should collaborate with researchers to develop curricula that integrate AI literacy frameworks. This need is in line with the issues identified by Lai and Tu (2024), which emphasize the importance of guaranteeing students' rights to learn GenAI, as well as the findings of Tu and Lu (2025), which show that the development and testing of GenAI as a teaching support tool is still minimal and needs to be prioritized. This is to meet diverse learning needs using GenAI with ethical considerations in higher education. A *prompt engineering* training program is also needed to help faculty maximize the potential of GenAI in learning practices through the *prompts* used with students.

Future research on prompt engineering should begin by exploring prompt engineering in non-English languages to determine how optimally GenAI performs on prompt inputs using non-English languages. Furthermore, a prompt engineering taxonomy and validated evaluation metrics need to be developed so that prompt engineering performance can be measured consistently and used for cross-testing across different types of tasks and disciplines. This facilitates the generalization of research findings. Prompt engineering is a very important skill for students because it allows them to utilize the full potential of GenAI and teaches AI literacy through understanding the principles of prompt engineering (Lo, 2023). Therefore, a research focus on prompt engineering could lead to the development of a curriculum integrated with GenAI. In the future, GenAI may become a common part of various aspects of life and serve as a foundation for students with 21st-century skills (Federiakin et al., 2024).

9. Conclusions

This scoping review has explored the potential of prompt engineering, which, based on preliminary evidence, indicates a positive association with the learning process using GenAI, especially in bridging student AI literacy in higher education. Based on an analysis of empirical studies that meet the inclusion criteria, the findings presented below are preliminary and should be interpreted in the context of a still limited corpus. This scoping review explores prompt engineering, starting from the explicit and implicit definitions of prompt engineering based on the included studies and the application of prompt engineering in various learning contexts in higher education, especially in English and computer science learning.

The selection of GenAI used to complete student assignments must be tailored to the needs, but in the studies reviewed, many chose ChatGPT to be applied in learning. The use of ChatGPT as a learning companion has a number of potential benefits, but there are also risks identified in the studies reviewed, such as a tendency toward dependence and the possibility of consuming inaccurate output. To prevent these risks, prompt strategies can be applied to direct GenAI output according to the context of the task and provide students with an understanding of AI literacy. The prompt strategies identified in the reviewed and reported studies associated with strengthening student AI literacy begin with assigning roles, formulating the desired context and task, setting boundaries, and requesting explanations or verification.

Preliminary evidence obtained from this review indicates that prompt engineering has the potential to introduce AI literacy to students based on the structure of the prompt strategy, which indirectly directs students' attention to the dimensions of AI literacy, namely awareness, use, evaluation, and ethics. Although this pattern was consistently identified in the studies reviewed, a stronger causal relationship requires confirmation through experimental research with larger samples and more rigorous designs. However, the application of prompt engineering to students still requires monitoring by lecturers through targeted learning designs to ensure that the prompt engineering application process is carried out correctly. Targeted learning designs integrate process-based assessments that can align learning incentives with learning objectives. The process evaluated by students is the use of prompt engineering in the log history of interactions with GenAI, including reasons for revisions and verification evidence for GenAI output. This approach is reported to motivate students to document the reasons behind their decisions and critically evaluate GenAI output.

CRediT authorship contribution statement

Fadlan Nugraha Nur Pangestu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Ida Hamidah:** Writing – review & editing, Validation, Methodology. **Lilis Widaningsih:** Writing – review & editing, Validation, Methodology. **Ade Gafar Abdullah:** Supervision, Conceptualization. **Nisa Aulia Saputra:** Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

Statement**:** During the preparation of this work, the author used DEEPL Write to improve readability and simplify the language and used ChatGPT as a tool to compile keywords relevant to the topic. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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